

Computing Systems, Networks and Digital Literacy								
	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can explore and operate familiar technology with support. I can recognise some technology that I use at home and in Nursery. I can use simple controls, such as buttons, switches and touchscreens, to make something happen. I can recognise that different types of technology can be used for different purposes.	I can recognise and name different types of technology used at home, at school and in the wider world. I can select and use familiar technology for a particular purpose. I can recognise that computers and other digital devices can help us to do different things. I can identify some of the main parts of a computer or digital device.	I can identify examples of technology in the classroom, school and wider world. I can explain that technology is something that has been created to help us do things. I can identify a computer and name its main parts. I can explain how the different parts of a computer allow us to interact with it.	I can identify examples of information technology at school, at home and in the wider world. I can explain some of the different ways information technology is used. I can explain some ways information technology benefits people.	I can explain that a digital device accepts an input, processes it and produces an output. I can identify examples of input, process and output in different digital devices. I can identify some of the physical components that make up a digital device. I can explain why digital devices may need to be connected to one another.	I can explain how computers and other devices are connected together to form networks. I can explain that networks can be connected to other networks. I can identify some of the devices involved in connecting networks. I can explain that the internet is a network of networks.	I can explain some of the ways computing technology has developed over time. I can identify significant developments that have contributed to modern computing. I can explain how computing technology has changed aspects of people's lives and society. I can explain why computing became particularly important during	I can explain how search engines help people find information on the World Wide Web. I can use appropriate search terms and queries to find information efficiently. I can explain that search engines use algorithms to select and rank results. I can explain why search results may appear in a particular order.

	I can begin to use technology carefully and responsibly. I can ask an adult for help when I am unsure about using technology.	I can use familiar digital devices with increasing independence. I can follow simple rules for using technology safely and responsibly. I can recognise that I should ask a trusted adult for help if something happens when using technology that I do not understand or like.	I can use a mouse or trackpad and keyboard to interact with a computer. I can explain why we have rules for using technology. I can follow agreed rules for using technology safely and responsibly. I can recognise when I should ask a trusted adult for help when using technology.	I can recognise that some technology can communicate with other technology. I can explain that the internet connects computers and other devices around the world. I can identify some of the ways people use the internet to find information and communicate. I can make choices about when using information technology is helpful or appropriate. I can explain how to use information technology and the internet safely and responsibly.	I can explain that a computer network connects computers and other digital devices. I can identify some of the physical components used to create a computer network. I can compare using a device independently with using a device connected to a network. I can explain some of the benefits of computer networks.	I can distinguish between the internet and the World Wide Web. I can explain that the World Wide Web is one of the services available through the internet. I can explain how websites and webpages can be accessed through the World Wide Web. I can recognise that digital content is created and added to the World Wide Web by people. I can recognise that online content has an owner and that other people's work should be used responsibly.	the Second World War. I can explain the purpose of the Enigma machine and why breaking the Enigma code was significant. I can explore the contributions of different people to the development of computing. I can recognise that contributions to the history of computing have come from people with a wide range of backgrounds and identities. I can make connections between developments in the history of computing and technology I use today. I can recognise that computing continues to develop and influence how people live, work	I can recognise that the highest-ranked search result is not necessarily the most useful or reliable. I can explain that search results and other online content can be influenced by commercial interests, including advertising. I can consider the reliability and relevance of information when selecting search results. I can explain different ways people communicate using digital technology.
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							and communicate.	I can compare different methods of online communication and select an appropriate method for a particular purpose or audience. I can evaluate some of the benefits and limitations of communicating using digital technology.
Vocabulary	Technology, device, button, switch, touchscreen, on, off, control, safe, help	Digital device, technology, computer, screen, keyboard, mouse, touchscreen, purpose, responsible, safe, trusted adult	Computer, technology, digital device, keyboard, mouse, trackpad, screen, input, responsible, rules, safe, trusted adult	Information technology, internet, digital device, information, communication, connected, network, benefit, appropriate, responsible, online	Network, input, process, output, digital device, component, connection, computer, switch, server, network cable, communicate	Internet, network, World Wide Web, website, webpage, router, server, content, browser, hyperlink, online, ownership	Computing, development, technology, code, Enigma, encryption, decryption, computer, innovation, communication, contribution, society	Search engine, search query, algorithm, ranking, search result, reliability, relevance, digital content, advertising, communication, audience, filter, website, World Wide Web

Creating Digital Content

	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can explore creating pictures using digital technology.	I can use a digital device to take photographs independently.	I can log in and access the digital tools and resources I need.	I can use a keyboard to compose and edit increasingly extended text.	I can use titles, headings, text, images and layout to communicate information effectively in a digital document.	I can design a digital presentation for a particular purpose and audience.	I can explain that vector graphics are created using individual digital objects and shapes.	I can analyse the structure and features of existing websites.
	I can use a touchscreen or simple controls to make marks and select tools.	I can make choices about what I include in a photograph.	I can use a mouse or trackpad and keyboard to create digital content.	I can format text by changing features such as font, size and colour.	I can create and edit a digital publication for a particular purpose.	I can select and use themes, layouts, text and images effectively within a presentation.	I can create a vector drawing by combining and manipulating shapes.	I can plan the structure, content and features of a webpage for a particular purpose and audience.
	I can take a photograph using a digital device with support.	I can use a computer or digital device to create a picture.	I can use a digital painting program and select appropriate tools to create a picture.	I can combine text and images to communicate information effectively.	I can make purposeful choices about layout and style.	I can edit and format digital images within a presentation.	I can use layers, objects and drawing tools to create a desired visual effect.	I can combine text, images and other digital content effectively within a webpage.
	I can use technology to listen to music, sounds and stories.	I can select different digital tools to change the appearance of my work.	I can compare creating artwork digitally with creating it using physical materials.	I can select and use an appropriate layout or template for a particular task.	I can explain how different layouts and styles may be appropriate for different purposes and audiences.	I can use animation and transitions purposefully rather than simply for decoration.	I can select and modify individual elements of a vector graphic without changing the whole image.	I can design a clear navigation path between webpages.
	I can use simple controls such as play, pause and stop.	I can use undo and redo to change something I have created.	I can use a computer to write and edit text.	I can save and retrieve digital content that I create.				
	I can explore how technology can be used to create and record different things.	I can use simple digital tools to create and combine pictures, marks and text.	I can add, remove and change text to improve my work.		I can combine different types of digital content within a publication.	I can make choices about how digital content should be presented to an audience.	I can independently design, create, evaluate and refine a vector graphic for a purpose.	I can preview, test and refine webpages before publishing them.

	I can talk about something I have created using technology.	I can save or retrieve digital work with support. I can talk about the choices I have made when creating digital content.	I can use a digital camera to take purposeful photographs. I can make simple changes to digital images, including applying filters. I can combine text and images to create a simple digital document for a purpose. I can explain some of the choices I have made when creating digital content.	I can review, edit and refine a digital document before publishing it. I can use a digital device to take purposeful photographs. I can explain some of the features that make a photograph effective. I can edit and enhance digital photographs for a particular purpose. I can explain how editing can change the appearance or meaning of a digital image.	I can explain that animation is created from a sequence of images or frames. I can plan a stop-motion animation using a sequence or storyboard. I can capture and organise images to create a stop-motion animation. I can edit an animation to improve its timing and effect. I can review and refine my digital content in response to its intended purpose.	I can explain that sound can be recorded and stored digitally. I can use a digital device to capture clear audio. I can edit digital audio to remove, rearrange or change recorded material. I can combine different audio elements, such as speech, music and sound effects. I can plan and create a digital audio production, such as a podcast, for a particular purpose and audience. I can evaluate my editing choices and refine the finished digital product.	I can identify the features of effective digital video. I can use a storyboard to plan the content and sequence of a video. I can capture purposeful video footage using a digital device. I can make decisions about framing, composition and the content I record. I can edit and combine video clips to create a coherent finished product. I can recognise when reshooting material would improve a video. I can evaluate and refine a video with reference to its purpose and audience.	I can explain that digital content has an owner and consider copyright when selecting images and other material. I can explain some of the implications of linking to or using content created by other people. I can create and manipulate three-dimensional digital objects. I can explain differences between creating graphics in two and three dimensions. I can construct a digital 3D model of a physical object. I can decompose a physical object into simpler three-dimensional shapes when planning a model.
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								I can design a digital model by combining and manipulating 3D objects. I can independently develop, evaluate and refine a digital 3D model for a particular purpose.
Vocabulary	Create, picture, photograph, camera, touchscreen, sound, play, pause, stop, digital	Digital content, camera, photograph, image, text, tool, select, undo, redo, save, create, edit	Digital content, keyboard, mouse, cursor, text, image, photograph, filter, document, edit, delete, insert, save, purpose	Format, document, font, layout, template, image, photograph, edit, enhance, crop, save, retrieve, publish, purpose	Audience, publication, layout, heading, style, animation, stop-motion, frame, sequence, storyboard, edit, timing, purpose, refine	Presentation, theme, transition, animation, audio, digital recording, podcast, sound effect, narration, edit, combine, audience, purpose, evaluate, refine	Vector, object, layer, graphic, manipulate, composition, video, footage, storyboard, framing, sequence, edit, reshoot, audience, evaluate, refine	Website, webpage, navigation, hyperlink, copyright, ownership, preview, digital content, 3D model, 3D object, manipulate, decompose, design, audience, evaluate, refine

Data and Information								
	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can sort objects into groups according to a simple feature.	I can sort objects into groups using a chosen feature.	I can label objects according to their properties.	I can collect data about objects and describe their attributes.	I can explain that a database is an organised collection of data.	I can explain that data can be collected over time to answer questions.	I can explain that a database stores organised collections of data.	I can identify questions that can be answered using data.
	I can recognise when objects are the same or different.	I can explain why an object belongs to a particular group.	I can group objects in different ways according to a chosen property.	I can use an attribute to sort and group objects.	I can identify attributes that can be used to describe and classify objects.	I can identify the data needed to answer a particular question.	I can identify fields and records within a database.	I can identify the data needed to investigate a question.
	I can count how many objects are in a group.	I can count and compare the number of objects in different groups.	I can count the number of objects within a group.	I can record data using a tally.	I can create yes/no questions that distinguish between objects.	I can explain that sensors can measure and record data about the physical world.	I can compare the advantages of storing data using paper and computer-based systems.	I can organise data effectively within a spreadsheet.
	I can use pictures or objects to show how many of something there are.	I can use technology to organise objects and information.	I can collect simple data to answer a question.	I can recognise that objects can be represented using pictures and other symbols.	I can organise yes/no questions into a branching database.		I can collect and record information consistently so that it can be stored within a database.	I can explain that spreadsheet cells can contain different types of data.
	I can use simple technology with support to organise or represent information.	I can represent simple data using pictures or a pictogram.	I can represent data using a pictogram.	I can create an accurate pictogram from collected data.	I can create and test a branching database away from a computer.	I can use a digital device to collect data automatically.		I can use formulae to calculate new data from existing values.
	I can talk about what I notice when looking at groups of objects.	I can use a simple pictogram to find out information.	I can use a pictogram to compare groups and answer questions.	I can use a pictogram to compare data and answer questions.	I can create a digital branching database.	I can explain that a data logger records data points from sensors over time.	I can group and sort data to help answer questions.	I can apply and duplicate formulae accurately across a spreadsheet.

		I can answer simple questions about data I have collected or represented.	I can explain how the data represented on a computer relates to the objects or information that were collected. I can recognise that the same data can provide information to answer different questions.	I can choose an appropriate way to organise or represent simple data. I can explain how a computer can be used to present information. I can use the information presented in a data representation to make a simple conclusion.	I can choose questions carefully so that a branching database identifies objects efficiently. I can use a branching database to identify an object from its attributes. I can test my database and make changes when it does not work as intended.	I can collect data over an appropriate period of time. I can examine data collected by a data logger and identify patterns or changes. I can use collected data as evidence when answering a question. I can recognise that the way data is collected can affect the conclusions that can be drawn from it.	I can use filters to select specific data from a database. I can combine criteria to investigate increasingly complex questions. I can use digital tools to compare and represent data visually. I can interrogate a database to answer real-world questions. I can interpret the results of a database search and explain what the data shows.	I can use a spreadsheet to model a real-world situation, such as planning an event. I can select an appropriate chart or other representation for the data I want to communicate. I can analyse data to identify patterns, relationships and anomalies. I can interpret data and use it to answer increasingly complex questions. I can evaluate whether the data available is sufficient to support a conclusion.
Vocabulary	Group, sort, same, different, count, object, picture,	Data, collect, sort, group, pictogram, information, compare, more,	Data, object, property, label, group, sort, collect, pictogram,	Attribute, data, tally, pictogram, collect, record, sort, group, compare,	Database, branching database, attribute, classify, yes/no question,	Data logger, sensor, data point, collect, measure, record, interval, pattern,	Database, field, record, filter, sort, criteria, query, interrogate, data, information,	Spreadsheet, cell, formula, calculated data, model, analyse, anomaly, pattern,

	information, more, fewer	fewer, question, represent	represent, compare, information, question	represent, information, conclusion	data, information, identify, organise, test, efficient	change, evidence, question, conclusion	compare, represent, interpret, results	relationship, chart, represent, interpret, data, conclusion, evaluate
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Programming and Computer Science

	Nursery (Understanding the World)	Reception (Understanding the World)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can follow simple instructions in the correct order. I can give a simple instruction to another person. I can explore what happens when I press buttons or use simple controls on a device. I can make choices about which button, icon or control to use. I can use simple directional language such as forwards, backwards, go and stop. I can notice when an action produces a particular result.	I can follow and give a sequence of simple instructions. I can put simple instructions into the correct order. I can use directional commands to control a simple device or program. I can predict what might happen when I give a command. I can create a short sequence of commands to achieve a simple goal. I can test my sequence and notice when it does not produce the result I expected.	I can explain that an algorithm is a precise sequence of instructions used to complete a task. I can follow and create simple algorithms. I can use commands to control a programmable device or onscreen object. I can combine commands into a sequence. I can predict the outcome of a sequence of commands. I can create a simple program from an algorithm. I can run and test a program to	I can explain the difference between an algorithm and a program. I can design an algorithm to achieve a particular outcome. I can create a program by turning an algorithm into a sequence of commands. I can use logical reasoning to predict the outcome of a program. I can explain how changing the order of commands can change the outcome. I can create programs in	I can explore a programming environment and identify how commands control a sprite. I can explain that a program has a starting point and that commands execute in a particular order. I can use sequencing to control the movement and actions of a sprite. I can recognise patterns within a sequence of commands. I can identify when a sequence of commands can be replaced by repetition. I can use loops to repeat commands	I can decompose a programming problem into smaller parts. I can design an algorithm before creating a program. I can use sequence and repetition together within a program. I can use count-controlled loops to repeat commands a specified number of times. I can use repetition to create patterns and shapes efficiently. I can predict the outcome of a program containing	I can explain that selection allows a program to make decisions. I can use conditional statements to control what happens when a condition is met. I can use if statements within a program. I can design, create and debug a program that uses selection. I can explain that a variable is a named value that can change while a program is running. I can identify everyday examples that help explain the concept of	I can design algorithms that combine sequence, repetition, selection and variables. I can use more than one variable within a program and explain how their values may change. I can predict how changing a variable or condition will affect the behaviour of a program. I can use logical reasoning to investigate and debug increasingly complex programs. I can decompose a complex programming

		I can change an instruction to make my sequence work more successfully.	check whether it works as intended. I can identify when a program does not work as expected. I can debug a simple program by finding and correcting an error. I can use logical reasoning to explain what I think a simple program will do.	which sequences of commands control actions or an animation. I can test a program systematically to identify errors. I can debug a program by changing commands or their sequence. I can compare different solutions to the same programming problem. I can evaluate a program and suggest how it could be improved.	within a program. I can explain how repetition can make a program shorter and easier to modify. I can design and create a program to meet a given purpose. I can test and debug a program containing sequences and repetition. I can modify an existing program and explain the effect of my changes. I can evaluate my program and refine it to improve the outcome.	repetition. I can identify and correct errors within programs containing loops. I can modify an existing program to produce a different outcome. I can compare different algorithms or programs that achieve the same outcome. I can explain why one programming solution may be more efficient than another. I can evaluate and refine my program against its intended purpose.	variables. I can create and modify variables within a program. I can use variables to store and change values within a program. I can design and create a program or game that combines sequencing, repetition, selection and variables. I can test how changes to conditions and variables affect the outcome of a program. I can evaluate and refine my program against its intended purpose.	problem into smaller manageable parts. I can explain that inputs provide data to a computer or controllable device. I can explain that outputs allow a computer or controllable device to produce a response. I can use user input to update a variable within a program. I can use conditional statements to compare a variable with a value and control the flow of a program. I can design and create a program that uses inputs, outputs, selection and variables to control a physical device. I can
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								systematically test and debug my program. I can evaluate the effectiveness and efficiency of my solution and refine it accordingly.
Vocabulary	Instruction, order, button, control, forwards, backwards, go, stop, action	Command, instruction, sequence, predict, direction, forwards, backwards, turn, test, change, outcome	Algorithm, program, command, sequence, code, predict, outcome, test, debug, error, logical reasoning	Program, algorithm, sequence, command, code, logical reasoning, predict, outcome, test, debug, error, evaluate	Repetition, loop, sequence, sprite, command, block, event, program, pattern, modify, debug, outcome, programming environment	Decomposition, repetition, count-controlled loop, algorithm, program, sequence, pattern, efficiency, predict, modify, debug, evaluate	Selection, condition, conditional statement, variable, value, if, sequence, repetition, input, program, debug, evaluate	Input, output, variable, selection, condition, sequence, repetition, decomposition, logical reasoning, physical computing, controllable device, program, debug, efficiency